

# LEONARDO URENA NIKOLOVA

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## PROFESSIONAL SUMMARY

Applied Data Scientist and Machine Learning Engineer with **5+ years of experience across production ML, AI research, and data-intensive engineering**, specializing in **entity resolution, embeddings, semantic similarity, classification, LLM applications, fine-tuning, and large-scale data integration**. Built production solutions across **200,000+ entity records, 50,000+ labeled classification examples, and 110,000+ identity records** using Python, SQL, PySpark, Databricks, MLflow, and Microsoft Azure.

Experienced defining **ground-truth datasets, baselines, evaluation metrics, model benchmarks, and error-analysis frameworks** to measure and improve ML and LLM systems. Strong background in supervised and unsupervised learning, deep learning, data normalization, record linkage, deduplication, model evaluation, production AI, and cross-functional technical decision-making.

## WORK EXPERIENCE

### Stryker

*Lead Cybersecurity Data Scientist*

Heredia, Costa Rica

*Oct 2023 – Present*

- Enabled **entity resolution and record linkage across 200,000+ records from 15 heterogeneous enterprise sources** by building Python, SQL, and PySpark pipelines for normalization, entity mapping, correlation, deduplication, reconciliation, and data validation.
- Reduced manual review volume by **55% while achieving a 96% true-negative rate** by developing and evaluating classification models trained on **50,000+ labeled historical reports**, including feature engineering, model comparison, validation, and systematic error analysis.
- Reduced preparation of **quarterly Board-level SOC security-posture reporting from weeks of manual analyst effort to minutes** by fine-tuning an **LLM on 2,000–3,000 analyst-reviewed historical ServiceNow Security Incident Records (SIRs)** to classify noisy, semi-structured investigation narratives into **MITRE ATT&CK tactics and Cyber Kill Chain stages**; validated performance on held-out closed incidents and analyzed confusion across adjacent taxonomy classes before operational use.
- Reduced average investigation time by **75%, from approximately 25 minutes to 6 minutes**, by architecting a production Agentic workflow using Python, Retrieval-Augmented Generation (RAG), Foundry SDK, Azure Functions, enterprise APIs, and LLM expert agents.
- Improved AI-output reliability by implementing an **LLM-as-a-Judge evaluation stage** that used a dedicated judge model to assess multi-agent analyses and recommendations against defined quality criteria before final outputs were released.
- Improved ML and LLM experimentation by creating **ground-truth benchmark datasets, baselines, quantitative metrics, model comparisons, and failure-mode analyses**, using evaluation results to guide model, retrieval, prompt, and workflow improvements.
- Improved retrieval relevance in production LLM applications by applying **embeddings, semantic similarity, semantic retrieval, and contextual grounding** to connect intelligent agents with relevant enterprise procedures and knowledge.
- Improved production classifier reliability by analyzing false positives, false negatives, edge cases, and labeled feedback; comparing model versions with **MLflow**; and validating updated models against benchmark datasets before operational adoption.
- Reduced identity and compliance review effort by **70% across 110,000+ identities in 61 countries** by correlating Active Directory, Workday, ServiceNow, endpoint, and policy datasets and identifying **1,200+ inconsistencies and exceptions**.
- Improved downstream analytical reliability by designing reusable **Python, SQL, PySpark, and Databricks pipelines** for ingestion, transformation, normalization, feature preparation, validation, correlation, and machine-learning workloads.
- Improved data quality across fragmented enterprise platforms by developing automated reconciliation workflows that identified **missing, duplicated, conflicting, and inconsistent entities** across heterogeneous data sources.
- Improved scalability and maintainability of production AI and ML systems by developing reusable backend and serverless components using **Azure Functions, REST APIs, Git, CI/CD, and version-controlled workflows**.
- Accelerated technical decision-making by communicating **model performance, experimental results, failure cases, data-quality findings, and technical trade-offs** to engineering, cybersecurity, infrastructure, incident-response, and business stakeholders in Agile/Scrum environments.

- Increased engineering efficiency by developing automation tools that replaced recurring manual processes and converted operational procedures into reusable technical workflows for global production environments.
- Improved reliability across complex Distributed Control System environments by investigating and resolving production issues involving distributed applications, infrastructure, operating systems, networking, and cybersecurity technologies.
- Improved integration and analysis of heterogeneous technical data by working across vulnerability management, SIEM, IDS, DLP, firewall, patch-management, Backup & Recovery, and ICS/OT threat-detection platforms.
- Increased consistency of security deployments by leading the design and implementation of a configurable hardening solution spanning operating systems, DCS infrastructure, historians, secure protocols, networking, ACLs, policy management, and Data Loss Prevention technologies.
- Improved resolution of customer and engineering issues by collaborating with international stakeholders to translate technical requirements, investigate system behavior, communicate findings, and deliver reliable production solutions.

## **AI & MACHINE LEARNING RESEARCH EXPERIENCE**

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**Artificial Intelligence for Natural Science – Tecnológico de Costa Rica**  
*Computer Researcher*

**Cartago, Costa Rica**  
*Jan 2021 – Jan 2022*

- Improved predictive performance by **23.5%**, reaching **89.93% final accuracy**, by designing, training, and evaluating a Deep Convolutional Neural Network for laboratory-validated lncRNA-miRNA interactions.
- Enabled similarity-based machine learning on complex biological sequences by creating **embedding-based numerical representations** of RNA data for predictive modeling and similarity analysis.
- Improved predictive feature quality by transforming long biological sequences and secondary-structure information into model-ready numerical representations through data preprocessing and feature engineering.
- Improved experimental reliability by constructing **laboratory-validated ground-truth datasets** and comparing predictive features and model configurations using accuracy, AUC, quantitative metrics, and systematic error analysis.
- Improved understanding of model limitations by analyzing prediction failures and ground-truth labels to identify recurring error patterns, evaluate predictive reliability, and guide iterative model improvements.
- Improved communication of experimental results by translating methodology, model performance, limitations, and technical findings into recommendations for multidisciplinary scientific collaborators.

## **TECHNICAL SKILLS**

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**Machine Learning & Data Science:** Python, SQL, Scikit-learn, TensorFlow, Keras, PyTorch, Machine Learning, Deep Learning, Classification, Supervised Learning, Unsupervised Learning, Feature Engineering, Exploratory Data Analysis, Model Training, Model Evaluation, Accuracy, AUC, Error Analysis

**Entity Matching & Data Quality:** Entity Resolution, Record Linkage, Entity Matching, Entity Mapping, Semantic Similarity, Similarity Analysis, Data Correlation, Data Deduplication, Data Normalization, Data Reconciliation, Data Validation, Data Quality

**Generative AI & LLM Engineering:** Generative AI, Large Language Models (LLMs), OpenAI Models, Fine-Tuning, Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, Grounding, Embeddings, Semantic Retrieval, LLM Applications, Expert-Agent Architectures

**ML & LLM Evaluation:** Ground-Truth Evaluation Datasets, Benchmark Datasets, Baseline Definition, Model Benchmarking, Evaluation Metrics, LLM Evaluation, LLM-as-a-Judge, Output-Quality Assessment, Model Comparison, Confusion-Matrix Analysis, Failure-Mode Analysis, Iterative Model Improvement

**Data Engineering:** PySpark, Spark, Databricks, Data Pipelines, Data Ingestion, ETL/ELT, Data Transformation, Data Integration, Data Preprocessing, Feature Preparation, API-Based Data Acquisition

**MLOps & Production AI:** MLflow, Experiment Tracking, Model Lifecycle Management, Model Validation, Model Maintenance, Production ML Workflows, Git, CI/CD, Version Control

**Cloud & AI Services:** Microsoft Azure, Azure Functions, Azure AI Foundry, Databricks, Cloud-Based AI Services, Cloud Infrastructure, Virtual Machines

**APIs & Automation:** REST APIs, API Integration, Azure Functions, Enterprise System Integration, Workflow Automation, AI-Assisted Automation, Backend Workflows

**Analytics & Visualization:** Power BI, Analytical Dashboards, Data Visualization, Operational Analytics, Data-Driven Decision Support

**Engineering Practices:** Agile, Scrum, Git, CI/CD, Cross-Functional Collaboration, Requirements Analysis, Technical Documentation, Stakeholder Communication

**Enterprise & Cybersecurity Platforms:** Workday, ServiceNow, Active Directory, Entra ID, EDR, SIEM, Vulnerability Management, Palo Alto Panorama, Forescout, Infoblox, SOAR

**Additional Programming:** Intermediate: Java, C++, MATLAB, Octave; Basic: Bash, PowerShell, JavaScript, HTML, CSS

**Languages:** English (C1), Spanish (Native), Russian (Native)

## EDUCATION

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**Tecnológico de Costa Rica**

Cartago, Costa Rica

*Licentiate Degree in Robotics and Mechatronic Engineering with Emphasis in Artificial Intelligence and Computer Vision*

**Instituto Jiménez Quesada**

Costa Rica

*Technician's Degree in Computer Support, Repair & Maintenance; Technician's Degree in Web Design*

## CERTIFICATIONS

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**Deep Learning Specialization** — DeepLearning.AI

**CompTIA Security+ Certification** — CompTIA